HealthCare Management System

Problem Statement:

Hospitals often face challenges in efficiently managing patient records, doctor schedules, lab tests, and appointments. The manual handling of these processes can lead to errors, delays, and inefficiencies, negatively impacting patient care and hospital operations. A streamlined, integrated system is needed to automate and centralize these tasks, ensuring better coordination among doctors, patients, and lab services while enhancing the overall quality of care.

Objectives of the Application:

1. **Centralized Patient Management:** Develop a comprehensive system for storing and managing patient information, including personal details, medical history, and appointment records.

2. **Efficient Appointment Scheduling:** Implement a robust scheduling module to facilitate the booking, updating, and canceling of appointments by patients, with real-time visibility for doctors and administrative staff.

3. **Doctor and Lab Coordination**: Create seamless communication channels between doctors and lab services, allowing for easy ordering of lab tests, tracking of lab orders, and access to lab reports.

4. **Role-Based Access Control**: Ensure secure access to the system by implementing role-based access control, where admins, doctors, and patients have specific permissions tailored to their roles.

5. **Scalability and Integration:** Design the system to be scalable, with the ability to integrate additional services or modules in the future, such as pharmacy management or billing services.

Microservices Overview:

The following are the microservices involved:

1. Doctor-Service
2. Patient-Service
3. Appointment-Service
4. Lab-Service
5. Auth-Service
6. EurekaServer
7. API-Gateway

User Stories:

Doctor-Service:

1. As an admin, I want to add doctors to the system so that their information is available for appointments.
2. As an admin, I want to delete doctors from the system so that only active doctors are available.
3. As an admin, I want to view all doctors to manage the doctor workforce.
4. As a doctor, I want to update my public profile information for patients to view.

Patient-Service:

1. As an admin, I want to add patients to the system so that their information is available for appointments.
2. As an admin, I want to delete patients from the system.
3. As an admin, I want to update patient's data which would be visible to the doctors.
4. As a patient, I want to view my profile.

Appointment-Service:

1. As a patient, I want to book appointment by selecting a doctor and booking available slot.
2. As a doctor, I want to view all appointments and would choose to update their status.
3. As an admin, I want to Generate Slots for each doctor for a specific month that will be available to book by patients.

Lab-Service:

1. As a doctor, I want to place lab orders, so that my patients can get necessary tests.
2. As a doctor, I want to generate lab reports, so that I can provide results to patients.
3. As a patient, I want to view my lab orders, so that I can know which tests I need to take.
4. As a patient, I want to view my lab reports, so that I can see my test results.
5. As an admin, I want to view all lab tests, so that I can manage lab operations.
6. As an admin, I want to add new lab tests, so that they are available for orders.

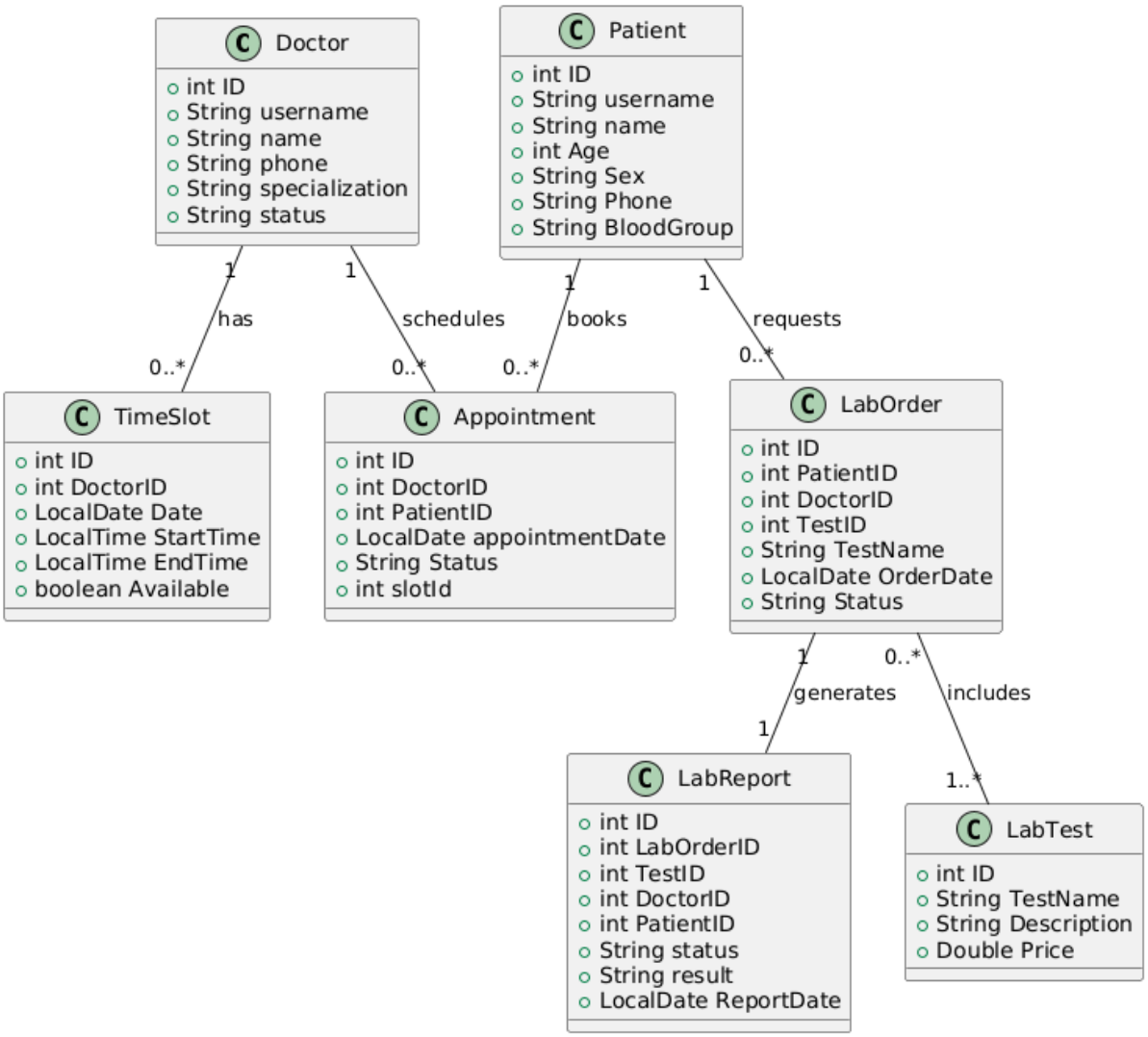
Auth-Service:

1. As an admin, I want to register accounts for doctors and patients for them to access their respective dashboards.
2. As a user with role ADMIN, DOCTOR or PATIENT, I want to login to the respective dashboards.
3. As a system, I want to validate JWT Tokens, so that I can ensure secure access to the services.

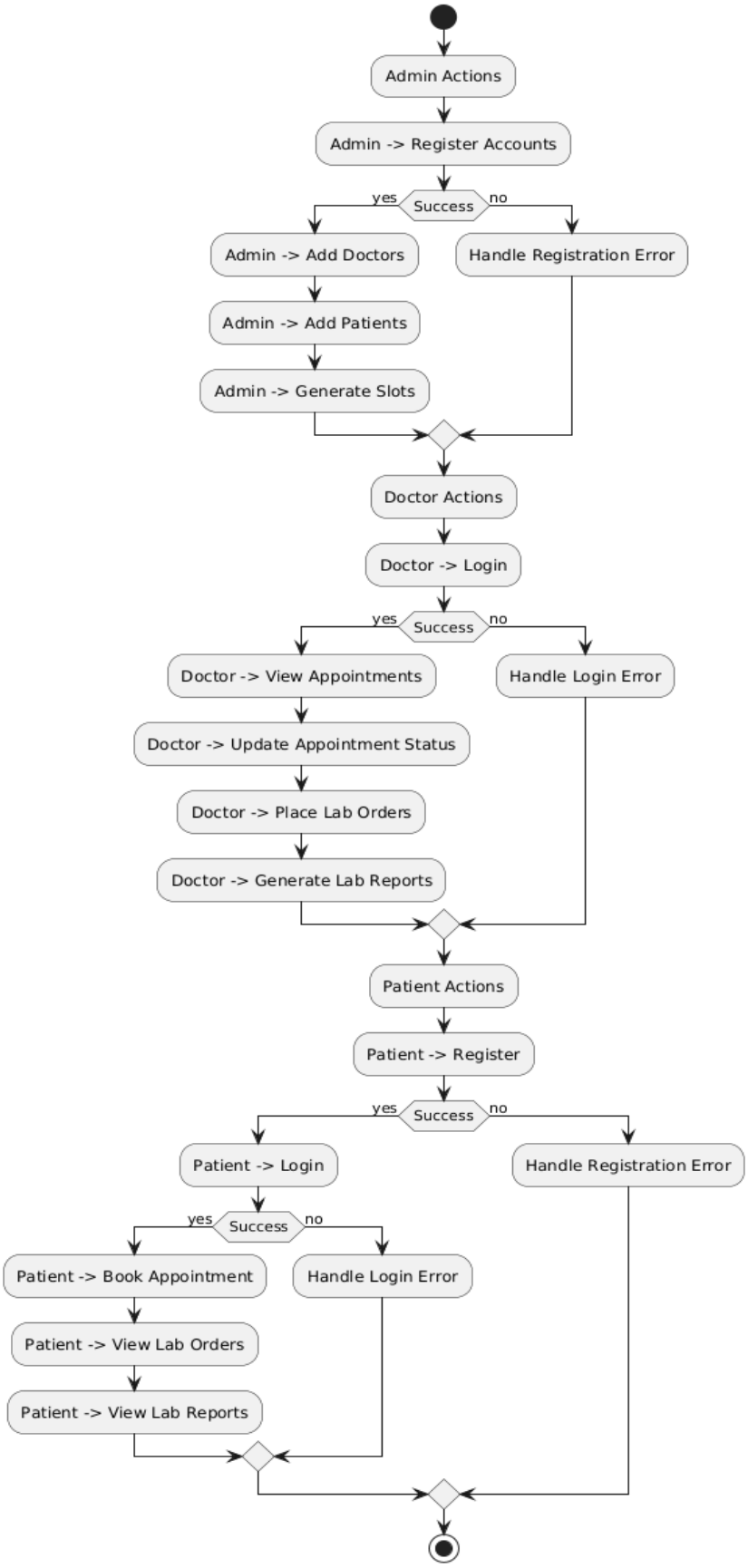
Eureka Server and API-Gateway:

1. As a developer, I want a Eureka Server for service discovery so that my microservices can find each other.
2. As a developer, I want an API gateway, so that I can route requests to the appropriate services.

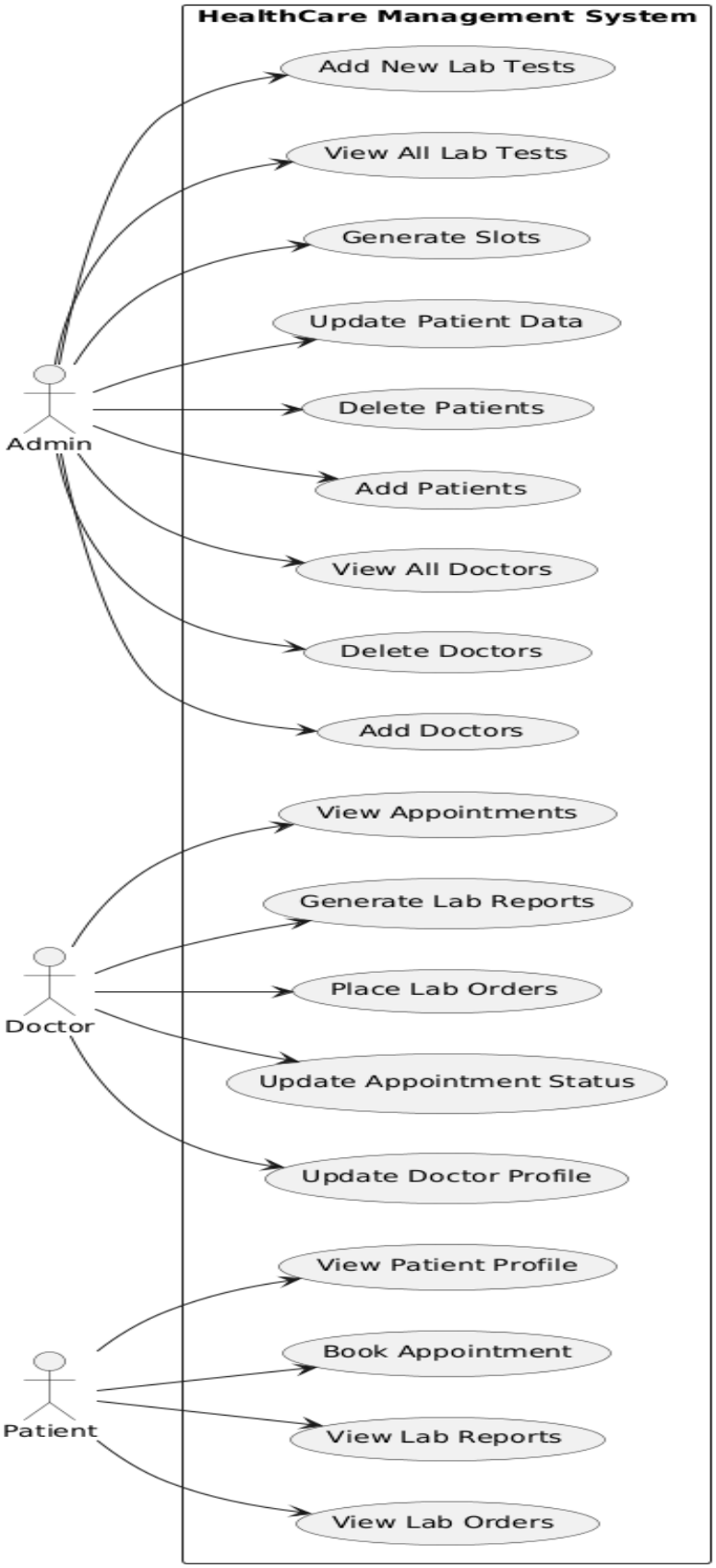
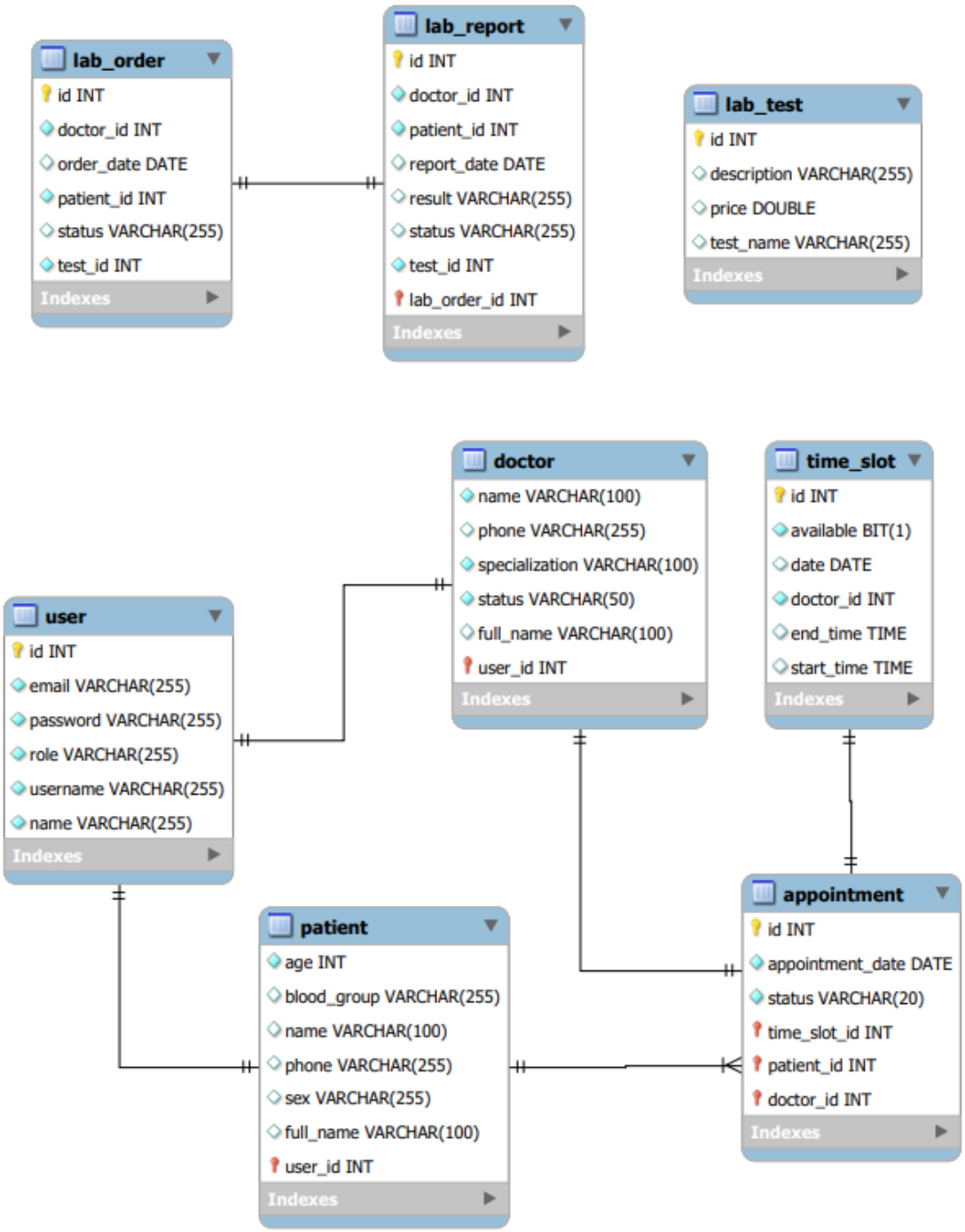
Class Diagram



Activity Diagram



Use Case Diagram



Entity Relationship Diagram